Lab 03 — The App

One thing today: add the four roles to your steps with your model key, read the first report, audit it by hand, and prove the desk on a company your partner picks.

You arrive with: your seven steps running on COST and WMT, pushed; your model API key in `.env` (a developer key, not a chat subscription); the report audit read; your gap list.

Open — the paired review (10 minutes)

1. **Tools alive.** Terminal in your work folder; your steps print the Costco facts; `check_env.py` prints the first four characters of your key; your AI agent answers.
2. **Defend.** Show your partner step 4 on WMT. Partner asks: which revenue tag, and why that one?
3. **Gap check.** Write one line: which of the four roles you expect to get a number wrong, and why.

D — the question, the same for everyone

> On the steps you built, can four model roles write a five-part report whose numbers survive an audit against the filings, and does the same desk hold up on a company you did not choose?

R — what you work from

Input	Where
Your steps and their saved folders	your repository
The roles and the template	Francis & Zhang (2026), Section 3.1, Figure 1, Figure 2

Input	Where
The checklist	The report audit: correct, wrong, unsupported, missing
Your model API key	in your `.env` from Lab 01
A company you did not choose	your partner picks it: any U.S. company that reported this month. Start here: EDGAR full-text search; the desk takes its ticker

I — build the app, then audit it

1. Send your AI coding agent this request word for word, filling the brackets:

> Now add the four roles of Francis & Zhang (2026), Figure 1, on top of my seven steps. Use the official Python SDK of [your model provider: for example OpenAI, Anthropic or Google], with my key read as API_KEY from .env. Each role is its own model call with its own instructions, and it receives only what the Strategy Manager sends it, as JSON with the task, the context and the expected output. The Research Analyst covers the business model, competitive position, catalysts and recent developments, from the 10-K's business and MD&A sections and the earnings release, and every claim quotes its line. The Financial Analyst does the statement analysis from step 4's facts and step 5's ratios, the quarter from step 6 and the peers from step 7; every number carries its filing and its line or tag and period, every ratio shows its arithmetic, and wherever a filing offers two definitions it says which one it used; it may ask the calculator for more ratios but never computes in the model. The Investment Advisor writes a thesis and Buy, Hold or Sell, with the one number that decides it, from the two roles' outputs and nothing else. The Strategy Manager checks that the five parts are present and consistent (Overview; Market Positioning and Growth Outlook; Valuation Insights; Future Developments and Strategic Recommendations; Investment Recommendation), sends any failing part back to its role once, then compiles the report. Print each role's name as it runs. Save the report as Markdown in the run's folder, with the filings read and their accession numbers on the first line and the run time on the second, and print its path. Nothing in the code names a company. Tell me the exact command to run the desk on a ticker with peers and a price.

Rewrite it in ASD-STE100 yourself before you send it, as Week 1 taught; a sample is here.

2. Run it: `COST --peers WMT,TGT,BJ --price P`, with Tuesday's close. *Expect:* seven steps, then four role names, then the path of a five-part report.
3. **While it runs:** read the report audit again, then work one item from your gap list in a second AI chat. Never collected.
4. **Audit, by hand.** Copy the audit table from the checklist into a new Markdown file in your repository. Fill it for your Costco report from the filings: the eleven core claims first, then every other number in the Financial Analyst's part, then the template and the recommendation rows. No AI for the marks: this is the human sign-off. Count the errors (wrong plus unsupported) and the missing claims, and write both at the top.
5. **The partner's pick.** Your partner names the company. Run the same command on its ticker, with peers you choose. *Expect:* a five-part report about that company, every number with its source, the latest quarter from the release it filed this month. Not running by 9:15? Use the fallback: it counts.

V — prove it on the known answer

Your Costco report states the eleven core claims, each with its source and each equal to what your Lab 02 steps printed and the filing shows; P/E on the close you gave it; and each peer's operating margin on the same definition as Costco's. Where it does not, your audit table says which role, and Week 3's prework fixes that role.

The partner's pick. Your partner checks two core claims of that report against its filings, by hand, and writes both marks at the top of the file. A report that quotes a line the filing does not contain is unsupported, however plausible. You audit all eleven yourself before Session 4.

The human sign-off — Learn on your own

1. What is it? Read the first screen of Anthropic's open-source financial-services agents, including the Earnings Reviewer: github.com/anthropics/financial-services. Find the sentence about human sign-off.
2. How does it work there: what do those agents refuse to do on their own?
3. Explain it: what does that repository's agent have that your desk lacked today?

Reflect

Explain to your partner: 1. which error a reader would have trusted most, and why; 2. which role you will change first, and what you will tell it.

Checkout — Checkpoint 1

By the end of the day (the schedule), in Brightspace → **Checkpoint 1**: a permalink (commit link) to your repository holding your desk and the report on your partner's pick (a fallback report counts).

Before Session 4 — do these: the list on the Week 3 page, about 45 minutes.

Next: Lab 04 rebuilds the desk as a team: the process a graph in code, the verifier a node no run can skip.